

INSPACK Ø=Þáp Ø=Üæ

AI-Based Legal Metrology Packaged Commodities Compliance & Inspection System

- **Problem Statement ID:**  
SIH-26034
- **Problem Statement Title:**  
Software System to check compliance of Packaged Commodities under the Legal Metrology (Packaged Commodities) Rules, 2011 by scanning products, images and labels
- **Theme:**  
Agriculture, FoodTech & Rural Development / Governance
- **PS Category:**  
Software (AI Multimodal Computer Vision + Deterministic Rules Engine)
- **Ministry / Organization:**  
Ministry of Consumer Affairs, Food & Public Distribution — Legal Metrology Division
- **Team Name (Registered):**  
Neural Knights (Ideas • Intelligence • Impact — Tech for a Fairer Market)
- **Institution / College:**  
Chaitanya Bharathi Institute of Technology (CBIT), Hyderabad (AICTE: 1-44641605560)
- **Live Demonstration Prototype:**  
<https://inspacknksih2026.vercel.app/>

OFFICIAL TEAM ROSTER

Chaitanya Bharathi Institute of Technology (CBIT)  
Nominated by Principal CBIT (Prof. C.V. Narasimhulu)

Ø=Üd Shaik Saleem (Team Leader)

Roll: 160124771129 | B.E AIDS-2, III Year

Ø=Üd D. Rishwanth Reddy

Roll: 160124771105 | B.E AIDS-2, III Year

Ø=Üd Dasari Hitharth

Roll: 160124771106 | B.E AIDS-2, III Year

Ø=Üd Pachimatla Hasini

Roll: 160124771084 | B.E AIDS-2, III Year

Ø=Üd B. Venkat Sai Ram

Roll: 160125733151 | B.E CSE-3, II Year

Ø=Üd G. Rohith Nandhan

Roll: 160125733158 | B.E CSE-3, II Year

AUTOMATED 5-STAGE STATUTORY INSPECTION WORKFLOW

1. Multi-View Capture:

Captures Front PDP, Back declarations, and Side batch panels via smartphone camera with guided viewfinder.
2. Legal Extraction:

Extracts 20+ statutory entities: Trade/generic name, declared quantity, SI units, MRP, dates, factory address, PIN.
3. Deterministic Rules:

Evaluates against LMPC Rules 6, 7 Table I, 10, 13, 18, 22, and Second Schedule standard pack sizes without AI hallucination.
4. Evidence Overlay:

Overlays interactive color-coded bounding boxes directly on package photos with exact gazette legal citations.
5. Statutory Dossier:

1-Click compilation of official Seventh Schedule Form A & Form B sheets with Section 65B electronic admissibility certificate.

CORE INNOVATION: EXPLAINABLE STATUTORY DECISION SUPPORT

- & Beyond Plain OCR

Existing OCR only extracts raw text into a box. INSPACK connects what is printed on the package to the exact legal requirement the inspector must verify.
- & Deterministic Laws

Aheads pixels, but statutory compliance is 100% deterministic. Rules, font heights, and compounding penalties are evaluated against gazette tables with zero hallucination.
- & Inspection-Support System

INSPACK does NOT replace the enforcement officer. It eliminates 20+ minutes of manual measuring and handwritten paperwork, giving the officer court-ready findings.

PROTOTYPE TEST SHOWCASE

Amul Taaza Toned Milk (1 L Pouch)

- Overall Status:

NON-COMPLIANT (Score: 72/100)
- Rule 7 Violation:

MRP numeral height is 2.2mm. Mandated minimum under Table I for >500ml is 4.0mm.
- Rule 6(2) Warning:

Toll-free helpline present, but mandatory consumer care email address is missing.
- Rule 32 Penalty:

Compounding penalty calculated: Rs. 2,000.
- Form Generated:

Seventh Schedule Form B (Volume Checking Data Sheet) ready with 1-click download.
- Evidence Standard:

Photographic exhibit recorded with SHA-256 digital verification hash under Section 65B.

V Technologies to be used (programming languages, frameworks, hardware) • Methodology and process for implementation

5-TIER LAYERED SYSTEM ARCHITECTURE

% Tier 1: Field Client Layer

Responsive Progressive Web App (PWA) & Mobile App with on-screen Principal Display Panel (PDP) viewfinder guides, torch toggle, and local offline cache.

% Tier 2: Pre-Processing & Quality

HTML5 Canvas luminance leveling, contrast stretching, and Laplacian blur gradient variance test (warns inspector before OCR if image sharpness is low).

% Tier 3: Vision & Entity Extraction

Dual Engine: (1) Prototype uses Google Gemini Multimodal Vision API; (2) Sovereign Production uses on-premise Document AI / OCR (TrOCR/LayoutLM) on NIC MeghRaj.

% Tier 4: Deterministic Rules Engine

Pure deterministic service evaluating Rules 6, 7 Table I, 10, 13, 18, 22, Second Schedule standard pack sizes, and First Schedule MPE error margins.

% Tier 5: Evidence & Dossier Layer

Client-side jsPDF compiles Seventh Schedule Form A (Weight) and Form B (Volume) data sheets with Section 65B electronic evidence certification and signature blocks.

TECHNOLOGY STACK MAPPING

- **Frontend Client:**  
React 19, TypeScript, Vite, Tailwind CSS v4
- **Edge / Offline OCR:**  
Tesseract.js Worker + HTML5 Canvas Pre-processing
- **Multimodal Vision:**  
Gemini Vision (Prototype) " Sovereign NIC AI (Final)
- **Statutory Dossier:**  
jsPDF + jsPDF-AutoTable (Official Form A & B)
- **Data Persistence:**  
IndexedDB / LocalStorage (Local-First) + Firestore

STATUTORY LEGAL RULES CODIFIED

- ' **Rule 6(1)(a-g):**  
Mandatory Name, Address, Net Qty, Dates, MRP, Batch
- ' **Rule 7 Table I:**  
Minimum numeral height based on net quantity & PDP area
- ' **Rule 13(4-5):**  
Strict SI units: g, kg, ml, l. Blocks "gm", "gms", "dozen"
- ' **Rule 18(5):**  
Detects unauthorized price stickers, smudging & dual pricing
- ' **Second Schedule:**  
Validates mandatory standard pack sizes across 19 categories
- ' **First Schedule:**  
Calculates Maximum Permissible Error (MPE) allowable deficiency

V Analysis of the feasibility of the idea • Potential challenges and risks • Strategies for overcoming these challenges

## TECHNICAL FEASIBILITY

- Runs on standard smartphones & tablets.
- Zero proprietary optical or laser hardware needed.
- Dual-engine: Cloud Vision + 100% offline edge OCR.
- Deterministic rule engine prevents AI hallucinations.

## OPERATIONAL FEASIBILITY

- Simple 3-step officer workflow: Aim → Scan → Doss
- On-screen bounding boxes enable instant visual review.
- Eliminates 20+ mins of manual measuring & paperwork.
- Outputs official Seventh Schedule Form A/B sheets.

## ECONOMIC & DEPLOYMENT VIABILITY

- Zero new field hardware expenditure (CapEx).
- Officers use existing government mobile devices.
- Phased rollout: State retail pilot → Pan-India scale.
- Massive cost & time savings across inspection zones.

## REAL-WORLD FIELD CHALLENGES & DEMONSTRATED MITIGATIONS

### & Poor Lighting & Blurry Photos:

Problem: Low retail lighting or shaky hands cause OCR failure. | Solution: Laplacian gradient sharpness filter tests image clarity on HTML5 canvas before running extraction. Warns officer if blurry.

### & Zero Network in Rural Mandis:

Problem: Field inspections occur in remote godowns without 4G/5G. | Solution: Client-side Tesseract.js Web Worker + LocalStorage vault enables 100% offline inspection and queue-based cloud sync.

### & Curved & Crinkled Packaging:

Problem: Labels on pouches and bottles wrap around packaging edges. | Solution: Multi-view panel acquisition (Front, Back, Side) segments text across multiple views, preserving label integrity.

### & Changing Gazette Rules & Fines:

Problem: Legal metrology rules and compounding fines evolve over time. | Solution: Dynamic Rule Configuration Engine allows departmental administrators to update fines, rules, and schedules without rewriting software.

V Potential impact on the target audience • Benefits of the solution (social, economic, environmental, etc.)

MULTI-STAKEHOLDER IMPACT ECOSYSTEM

Ø=Ün FOR LEGAL METROLOGY OFFICERS

- Reduces inspection time from 25 mins to under 60 seconds.
- Eliminates manual letter-height calipers & handwritten forms.
- Provides court-ready Seventh Schedule Form A & B dossiers.

Ø=PÒ FOR CONSUMERS & CITIZENS

- Protects against stealth shrinkflation & non-standard pack sizes.
- Eliminates unauthorized sticker price hikes & hidden dual pricing.
- 1-tap grievance filing to National Consumer Helpline (NCH 1915).

Ø<ßí FOR FMCG BRANDS & PACKERS

- Brand Pre-Check Portal audits artwork before printing wrappers.
- Prevents expensive market product seizures and packaging recalls.
- Promotes transparent, standardized compliance across the supply chain.

Ø<BÜp FOR MINISTRY & DIRECTORATE

- National Surveillance Hub aggregates state-wise violation trends.
- Identifies repeat offender brands and high-risk commodity classes.
- Empowers data-driven policy amendments with empirical evidence.

WORKFLOW COMPARISON: MANUAL VS. INSPACK

%È Information Capture:

Manual: Manual measurement with magnifying glass & ruler

INSPACK: Automated multi-panel photo capture (Front, Back, Side)

%È Rule Verification:

Manual: Cross-referencing multiple gazette books & tables manually

INSPACK: Instant deterministic rule validation across all LMPC clauses

%È Numeral Height Check:

Manual: Manual measurement with plastic calipers (Subjective)

INSPACK: Automated comparison against Table I minimum standards

%È Tampered Price Check:

Manual: Hard to detect subtle sticker alterations in dim lighting

INSPACK: Automated detection of glued price stickers & dual pricing (Rule 18)

%È Pack Size Audit:

Manual: Officer memorizes allowed sizes across 19 categories

INSPACK: Instant validation against Second Schedule standard pack sizes

%È Report Preparation:

Manual: Handwritten Form A/B sheets & seizure memos (20-30 mins)

INSPACK: Instant 1-click Seventh Schedule Form A/B PDF (< 60 secs)

Ø=ÜÜ STATUTORY ACTS & LEGISLATION

- **The Legal Metrology Act, 2009 (Act No. 1 of 2010):**  
Statutory authority for powers of inspection, search & seizure (Sec 15), non-standard package penalties (Sec 36), and compounding of offences (Sec 48).
- **LMPC Rules, 2011 (Gazette of India):**  
Comprehensive statutory code prescribing mandatory package declarations, Principal Display Panel geometry, and enforcement schedules.
- **GSR 779(E) & 2021 Amendments:**  
Mandated statutory Unit Sale Price (USP) declarations and e-commerce digital display requirements on dark store platforms.

Ø=ÜÑ SCHEDULES & PRESCRIBED FORMS

- **Seventh Schedule [Rule 19(2)] - Form A:**  
Official statutory format for Weight Checking Data Sheet, tare calculation, and tripartite signatures.
- **Seventh Schedule [Rule 19(2)] - Form B:**  
Official statutory format for Volume and Measure Checking Data Sheet for liquid/fluid commodities.
- **Second Schedule [Rule 5]:**  
Mandatory standard packaging quantities across 19 commodity classes (Biscuits, Edible Oils, Tea, Rice, etc.).
- **First Schedule [Rules 2(e) & 22]:**  
Maximum Permissible Error (MPE) allowable deficiency tolerance tables.

&-p ELECTRONIC EVIDENCE & ADMISSIBILITY

- **Section 65B Indian Evidence Act / BSA 2023:**  
Prescribes statutory certification for electronic records, cryptographic SHA-256 hash preservation, and tamper-evident image logs.
- **W3C Canvas & Web Worker Standards:**  
Authoritative open web standards powering 100% offline, on-device image contrast optimization and edge OCR processing.
- **FSSAI Packaging Regulations, 2018:**  
Cross-referenced for front-of-pack synthetic color and artificial sweetener statutory warnings.

Ø<B HACKATHON ALIGNMENT & LIVE PROTOTYPE

- **SIH 2026 Problem Statement ID:**  
SIH-26034 (Ministry of Consumer Affairs, Food & Public Distribution — Legal Metrology Division).
- **Functional Live Web Prototype:**  
<https://inspacknksih2026.vercel.app/> (Tested with benchmark retail commodities).
- **SIH Implementation Guidelines:**  
Grounded in government deployment maturity, open-source compliance, and sovereign data privacy standards.